

**HANOI UNIVERSITY OF
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**SCHOOL OF INFORMATION AND COMMUNICATION
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DATA VISUALIZATION REPORT

Telecom Customer Churn Analysis

Dataset, Exploratory Analysis, and Visualization Technique Plan

Course: Data Visualization

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Abstract

This report presents our customer churn analysis workflow, from the public Kaggle dataset to exploratory analysis, feature grouping, model-assisted retention prioritization, and the final Power BI dashboard. The dataset was downloaded from Kaggle and split into churned and non-churned customer files for analysis. Following the course capstone guideline, we document the dataset source, churn-rate interpretation, Random Forest workflow, and the visualization techniques used in the final dashboard screenshots. The dashboard prioritizes clear denominators, readable comparisons, a colorblind-aware palette, and interactive risk exploration over adding every possible chart type.

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1 Introduction

1.1 Business Context

Customer churn is a critical issue for subscription-based telecom companies. When customers leave, the business loses recurring revenue and may need to spend additional resources to acquire replacement customers. We use the churn report to identify high-risk customer groups and to explain which customer attributes are associated with a higher churn rate.

1.2 Project Objectives

The project has four objectives:

1. Analyze customer churn patterns across customer, account, service, and experience dimensions.
2. Compare churned and non-churned customers through exploratory data analysis.
3. Build interpretable feature groups that support churn-rate comparison.
4. Explain how course visualization techniques are applied to the final dashboard pages and summarize retention recommendations based on the observed signals.

1.3 Current Scope

This Overleaf-compatible report contains the cover page, table of contents, dataset source, dataset statistics, EDA plan, model workflow, expected insight narrative, and a technique-application section aligned with guideline.md. The final dashboard screenshots come from the group's reporting artifact in the `img/` folder. Because the dashboard was finished under a fixed submission timeline, we focus the report on the techniques that are actually visible in the final pages and keep remaining improvements in the limitations section.

2 Dataset

2.1 Dataset Source

The dataset is a large customer churn prediction dataset with simulated, privacy-preserving customer records. It is available on Kaggle and can be accessed at: <https://www.kaggle.com/datasets/isandeep06/customer-churn-prediction-dataset-1m/data>. It can also be searched through Google Dataset Search: <https://datasetsearch.research.google.com/search?query=Customer%20Churn%20Prediction%20Dataset%201M>.

The source-level statistics are summarized in Table 1.

Table 1: Kaggle source listing summary

Item	Value
Dataset title	Customer Churn Prediction Dataset 1M
Source platform	Kaggle
Original scale	1,000,000 customer records
Original feature count	28 features + 1 target variable
Main target	Customer churn label

Table 2 summarizes all variables included in the dataset, grouped into their respective categories along with a brief description of each feature.

Table 2: Dataset feature description

Category	Column	Description
Customer Demographics	customer_id	Unique customer identifier
	signup_date	Customer registration date
	age	Customer age (18–90)
	gender	Gender: Male, Female, Other
	annual_income	Annual income in USD
	education	Education level
	marital_status	Marital status
	dependents	Number of dependents (0–5)
	senior_citizen	1 if age ≥ 65 , else 0
Account Information	tenure	Months with company (1–72)
	contract	Contract type: month_to_month, one_year, two_year
	payment_method	Payment method
	paperless_billing	Paperless billing: Yes/No
Service Usage & Billing	monthlycharges	Monthly service charges (\$20–\$200)
	totalcharges	Total charges to date
	num_services	Number of subscribed services (1–6)
	has_phone_service	1 if has phone service
	has_internet_service	1 if has internet service
	has_online_security	1 if has online security
	has_online_backup	1 if has online backup
	has_device_protection	1 if has device protection
	has_tech_support	1 if has tech support
	has_streaming_tv	1 if has streaming TV
	has_streaming_movies	1 if has streaming movies
Customer Behavior & Risk	customer_satisfaction	Satisfaction score (1–10)
	num_complaints	Complaints in last year (0–8)
	num_service_calls	Service calls last month (0–12)
	late_payments	Late payments last 3 months (0–5)
	avg_monthly_gb	Average monthly data usage (GB)
	days_since_last_interaction	Days since last contact (1–365)
Target Variable	credit_score	Credit score (300–850)
	churn	Target: 1 if churned, 0 if retained

2.2 Preprocessing

The dataset was cleaned by removing records with missing values and split into churn and non-churn subsets for the group analysis.

The resulting datasets' sizes are as shown in Table 3.

2.3 Working Dataset Descriptive Statistics

After cleaning and splitting the downloaded data, the working dataset used in this report contains 840,765 rows and 32 columns, with no missing values. Numeric descriptive statistics are shown in Table 4.

Table 3: Current dataset size

Segment	Number of customers
Churn	83,362
Not churn	757,403
Total	840,765

Table 4: Numeric statistics in the working dataset

Feature	Mean	Median	Min	Max
Age	44.72	44.00	18.00	90.00
Annual income	58,809.80	48,973.70	20,000.00	250,000.00
Dependents	0.80	1.00	0.00	5.00
Tenure	22.40	16.00	1.00	72.00
Senior citizen	0.20	0.00	0.00	1.00
Monthly charges	86.43	85.46	20.00	854.96
Total charges	1,838.74	1,250.23	16.30	16,252.86
Num services	2.56	2.00	1.00	6.00
Has phone service	0.77	1.00	0.00	1.00
Has internet service	0.85	1.00	0.00	1.00
Has online security	0.34	0.00	0.00	1.00
Has online backup	0.43	0.00	0.00	1.00
Has device protection	0.30	0.00	0.00	1.00
Has tech support	0.50	0.00	0.00	1.00
Has streaming TV	0.60	1.00	0.00	1.00
Has streaming movies	0.55	1.00	0.00	1.00
Customer satisfaction	6.16	7.00	1.00	9.00
Number of complaints	0.70	1.00	0.00	7.00
Number of service calls	1.76	1.00	0.00	12.00
Late payments	0.40	0.00	0.00	5.00
Average monthly GB	39.08	27.75	0.00	557.82
Days since last interaction	44.47	31.00	1.00	365.00
Credit score	678.54	680.00	300.00	850.00
Churn	0.10	0.00	0.00	1.00

Additionally, table 3 shows that the baseline churn rate is approximately 9.92%. This number is used as the reference point when interpreting group-level churn rates. A group above this baseline should be considered higher risk, while a group below the baseline is relatively lower risk.

3 Feature Selection and Grouping

3.1 Rationale

The dataset contains a diverse set of demographic, account, service usage, and customer behavior variables that may influence customer churn. To facilitate a structured exploratory data analysis, features are organized into logical categories based on their business context. This categorization improves the interpretability of the analysis by allowing related variables to be examined together and helps identify patterns associated with customer retention and churn. In addition, selected continuous variables are grouped into meaningful ranges where appropriate to simplify visualization and discussion of customer characteristics. The same feature organization is maintained throughout the project to ensure consistency between the

exploratory analysis and the predictive modeling stages.

3.2 Selected Overall Model Features

The overall churn prediction model uses the full dataset with both churned and non-churned customers. The selected raw features are:

- tenure
- contract
- customer_satisfaction
- num_complaints
- num_service_calls
- late_payments
- monthlycharges
- num_services
- has_tech_support

These variables cover four main analytical dimensions: customer tenure, contract commitment, customer experience, and service/payment behavior.

3.3 Correlation Matrix Screening

Before finalizing the Phase 2 feature set, the encoded overall dataset was screened with a Pearson correlation matrix. The full matrix is exported by the pipeline as `overall_encoded_correlation_matrix.csv`. Because the full matrix is too wide for the report, Table 5 summarizes the strongest encoded-feature correlations with the churn target using shortened feature labels. These values should be read as univariate screening signals, not as final causal evidence.

Table 5: Top encoded features by absolute correlation with churn

Feature signal	Corr.	Corr.	Interpretation
contract: two-year	-0.123686	0.123686	Two-year contracts are associated with lower churn.
contract: one-year	0.100979	0.100979	One-year contracts have higher churn than the two-year group.
satisfaction: 1-3	0.080981	0.080981	Low satisfaction is associated with higher churn.
contract: month-to-month	0.078836	0.078836	Month-to-month contracts are associated with higher churn.
complaints: <= 1	-0.066236	0.066236	Fewer complaints are associated with lower churn.
complaints: > 1	0.066236	0.066236	More complaints are associated with higher churn.
satisfaction: 7-9	-0.063678	0.063678	High satisfaction is associated with lower churn.
service calls: <= 3	-0.060627	0.060627	Fewer service calls are associated with lower churn.
service calls: > 3	0.060627	0.060627	More service calls are associated with higher churn.
has tech support	-0.045187	0.045187	Technical support is associated with lower churn.

At the feature-family level, the largest absolute correlations are concentrated in contract type, customer satisfaction, complaints, service calls, technical support, and late payments. Tenure and monthly charges are much weaker in this screening. This supports using the customer-experience and contract-related features as the primary explanatory features in the dashboard, while treating low-correlation variables as supporting context rather than dominant churn drivers.

Table 6: Feature-family correlation summary

Feature family	Max corr.	Mean corr.	Top encoded feature
contract	0.123686	0.101167	contract_two_year
customer_satisfaction_group	0.080981	0.049728	satisfaction_1_3
num_complaints_group	0.066236	0.066236	complaints_le_1
num_service_calls_group	0.060627	0.060627	service_calls_le_3
has_tech_support	0.045187	0.045187	has_tech_support
late_payments_group	0.041126	0.033550	late_0
num_services_group	0.027798	0.021552	services_1_2
tenure_group	0.011551	0.005833	tenure_32_plus
monthlycharges_group	0.000050	0.000050	monthly_ge_200

The churn-only dataset is not used for target-correlation screening because it contains only churned customers. In that file, the churn target has no class variation, so correlation with churn is undefined. Churn-only correlations are therefore useful only for profiling relationships inside the churned-customer group.

3.4 Selected Churn-Only Support Features

The churn-only file contains only customers who already churned, so it is not used to train a standalone binary classifier. Instead, it supports profiling and explanation of the churned customer group. The selected support features are:

- annual_income
- education
- contract
- paperless_billing
- num_complaints
- num_service_calls
- late_payments
- monthlycharges

3.5 Feature Grouping Rules

Table 7: Raw features and grouping rules used by the project pipeline

Raw feature	Grouped feature	Grouping rule
annual_income	annual_income_group	< 30K, 30–60K, 60–90K, > 90K
education	education_group	master/PhD = graduate; others = non-graduate
tenure	tenure_group	1–6, 7–15, 16–31, 32+ months
contract	contract	original contract category, one-hot encoded for modeling
paperless_billing	paperless_billing_flag	No = 0; Yes = 1
customer_satisfaction	customer_satisfaction_group	1–3, 4–6, 7–9
num_complaints	num_complaints_group	≤ 1, > 1
num_service_calls	num_service_calls_group	≤ 3, > 3
late_payments	late_payments_group	0, 1–2, 3+
monthlycharges	monthlycharges_group	< 200, ≥ 200

Raw feature	Grouped feature	Grouping rule
num_services	num_services_group	1–2, 3–4, 5–6
has_tech_support	has_tech_support	binary 0/1 feature

3.6 Interpretability Benefit

The selected features are not only model inputs; they also provide a structure for explaining churn. For example, satisfaction and complaints describe customer experience, contract and tenure describe commitment, and service/payment features describe operational behavior. This organization makes the final analysis easier to communicate to non-technical audiences.

4 Prediction Workflow

4.1 Random Forest Pipeline

The pipeline uses the cleaned churn dataset, creates grouped features, encodes categorical variables, trains a Random Forest classifier, and exports model outputs for later analysis.

4.2 Model Evaluation Summary

Table 8 reports the Random Forest test metrics:

Table 8: Random Forest test metrics from the group repository

Metric	Value
ROC AUC	0.676510
Accuracy	0.761694
Precision	0.187745
Recall	0.421945
F1-score	0.259863

4.3 Model Performance Interpretation

The model performance should be interpreted in the context of churn prediction, rather than as a generic classification score. In the working dataset, the number of churned customers is 83,362 out of 840,765 customers, which gives a baseline churn rate of approximately 9.92%. This means that the target class is strongly imbalanced: only about 10 out of every 100 customers are churned customers.

For this reason, accuracy is not the primary evaluation metric. A naive model that predicts every customer as non-churn would already obtain about 90.08% accuracy, but it would detect no churned customers. The Random Forest accuracy of 76.17% should therefore not be interpreted as the main indicator of model quality. More informative metrics are precision, recall, F1-score, ROC AUC, and precision-recall based metrics. In this run, the recall of 42.19% means that the model identifies about 42% of the actual churned customers, while the low precision reflects the difficulty of separating a relatively small churn class from a much larger non-churn class.

The model is also constrained by the Phase 2 feature design. The initial model uses interpretable, dashboard-friendly grouped features such as contract type, customer satisfaction group, complaint group,

service-call group, late-payment group, monthly-charge group, tenure group, service-count group, and technical support. These variables contain useful directional signals, but they do not capture all possible churn drivers. For example, the dataset does not include detailed behavioral histories such as plan changes, discount history, competitor offers, support ticket text, customer sentiment, or detailed payment timelines.

Another limitation is that several numeric variables are grouped into bins before modeling. This helps the dashboard remain readable and supports clear segment-level interpretation, but it can remove continuous detail that a predictive model might otherwise use. Therefore, the Random Forest model should be treated as a decision-support component for prioritizing retention actions, not as a perfect churn prediction system.

The selected threshold is chosen on the validation set to improve the balance between precision and recall. Lowering the threshold would identify more churned customers but would also increase false positives. Raising the threshold would improve precision but miss more churned customers. This trade-off is acceptable for a retention dashboard, where the objective is to highlight customers or segments that deserve earlier attention from customer support or retention teams.

Future model improvement should focus on using raw numeric features together with grouped features, testing gradient boosting models such as XGBoost, LightGBM, or CatBoost, tuning hyperparameters against F1 or PR-AUC rather than accuracy, handling class imbalance with class weights or resampling inside the training pipeline, and adding richer behavioral features if they become available.

5 Exploratory Data Analysis Plan

5.1 How Churn-Rate Charts Should Be Read

In churn-rate charts, a bar does not represent a distribution that sums to 100% across groups. Each bar represents the churn rate inside one group:

$$\text{Churn Rate}_{\text{group}} = \frac{\text{Number of churned customers in the group}}{\text{Total customers in the group}} \times 100\%.$$

This means each group should be compared independently against the baseline churn rate of approximately 9.92%.

5.2 Customer Experience Features

Table 9: Planned EDA interpretation by feature

Feature	Preliminary interpretation
Customer satisfaction	Higher satisfaction tends to lean toward non-churn. Low satisfaction groups are expected to have higher churn risk.
Number of complaints	More complaints are associated with a higher churn rate and should be treated as a strong customer-experience risk signal.
Number of service calls	More service calls may indicate unresolved service problems and higher churn risk.
Late payments	More late payments are associated with higher churn rate, although the signal is weaker than satisfaction and complaints.
Average monthly GB	Usage groups have relatively similar churn rates; this feature is weak as a standalone churn driver.
Days since last interaction	Churn rates are expected to be nearly flat across time-since-interaction groups, so this feature should be combined with other signals.

Feature	Preliminary interpretation
Credit score	Credit-score groups show small churn-rate differences and are not expected to be a major standalone churn signal.

5.3 Planned EDA Visuals

For each selected feature, the final report should include two complementary visuals:

1. Customer count by churn status for each group.
2. Churn rate for each group.

This structure separates customer distribution from churn risk. A large segment does not automatically mean high churn risk, and a small segment can still be important if its churn rate is much higher than the baseline.

5.4 Priority of Feature Interpretation

The feature interpretation should be ordered by actionability. Customer experience and payment-behavior variables are more useful for retention planning because they can trigger operational actions. Usage and credit variables should be treated as supporting context unless they show a clear above-baseline churn-rate pattern.

1. Customer satisfaction.
2. Number of complaints.
3. Number of service calls.
4. Late payments.
5. Average monthly GB.
6. Days since last interaction.
7. Credit score.

6 Visualization Design Principles for Churn Analysis

6.1 Audience and Analytical Questions

We designed the visualization artifact for analysts and customer-retention stakeholders. The dashboard does not only report how many customers churned; it also points to customer groups above the baseline churn rate and helps the team decide which groups deserve retention attention.

The core analytical questions are:

1. What is the overall churn baseline?
2. Which customer segments are large enough to matter operationally?
3. Which segments have churn rates above the baseline?
4. Which features are actionable for a retention team?
5. How do model outputs support, but not replace, business interpretation?

6.2 Metric Semantics

We separate distribution metrics from risk metrics throughout the report. This distinction matters because a segment can contain many customers but still have an average churn rate, while a smaller segment can have a much higher churn rate.

Table 10: Metric definitions used in the visualization design

Metric	Definition	Interpretation rule
Customer count	Number of customers in a segment.	Measures segment size, not churn risk.
Churn count	Number of churned customers in a segment.	Useful for workload sizing, but affected by segment size.
Churn rate	Churn count divided by all customers in the same segment.	Measures within-segment risk and is compared with the 9.92% baseline.
Share of churned customers	Churned customers in a segment divided by all churned customers.	Shows composition of churned customers only; values can sum to 100%.
Predicted churn probability	Model-estimated probability of churn for customers or groups.	Useful for prioritization, not as a final business decision by itself.
Average predicted churn probability	Mean predicted churn probability within a selected category.	Compares category risk; values are averaged and are not expected to sum to 100%.
Risk level	Grouped model or rule-based risk category.	Uses consistent thresholds and colors across pages.

6.3 Chart Selection by Analytical Task

We chose chart types by task rather than by visual style. The churn dataset is mainly table data, so the dashboard relies on bar charts, distribution views, relationship views, and summary tables.

Table 11: Chart types used for churn analysis tasks

Task	Recommended chart	Churn example
Compare counts	Grouped bar chart	Churn vs. non-churn by satisfaction group
Compare risk	Bar chart with baseline	Churn rate by complaint group
Show distribution	Histogram or box plot	Credit score or monthly charges
Show composition	100% stacked bar	Share of churned customers by segment
Show relationship	Scatter, heatmap, or small multiples	Satisfaction vs. complaints
Show model behavior	Metric cards and confusion matrix	AUC, precision, recall, F1-score
Compare predicted risk dynamically	Horizontal bar chart with a single-select feature slicer	Average predicted churn probability by category within the selected feature

As a team, we chose horizontal bar charts for most category comparisons because many labels are long, such as *Number of Service Calls*, *Customer Satisfaction*, and *Month-to-month*. Horizontal bars keep these labels readable while still using bar length for comparison.

6.4 Visual Encoding Rules

In the final dashboard, we rely on encodings that are easy to decode during a short presentation:

- Position and bar length compare counts, churn rates, and feature importance.
- Color hue separates categorical states such as churn versus non-churn or low/medium/high risk.
- Baseline lines on churn-rate charts make above-baseline groups easy to detect.
- We avoid pie charts, donut charts, and 3D charts where accurate comparison matters.
- Color is not the only carrier of meaning; we pair color with labels, ordering, or reference lines.

We keep the 9.92% overall churn rate as the main reference point because it prevents a common reading mistake: a large customer segment is not necessarily a high-risk segment. The baseline lets viewers compare each group by risk rather than by size alone.

6.5 Color System

The final dashboard uses a gray–orange–blue direction rather than a red–green palette. This was a team decision after reviewing the dashboard screenshots: orange is visible as a warning color, blue works as the main comparison color, and gray keeps supporting context quiet. The palette is more readable for color-vision-deficient users than red–green contrast, and we still support the colors with labels, sorting, and titles so the interpretation does not depend on hue alone.

Table 12: Accessible semantic color palette

Meaning	Color role	Suggested hex
High risk / warning	Orange highlight	#D55E00
Selected or primary comparison	Blue main signal	#56B4E9
Neutral, low emphasis, or context	Gray support color	#7A7A7A
Not churn / inactive context	Muted gray-blue	#64748B
Background	Page background	#F8FAFC
Gridline / border	Low-contrast structure	#E5E7EB

These colors remain stable across pages. Orange is reserved for warning or highest-risk emphasis; blue carries the main selected comparison; gray provides context. When color highlights a high-risk segment, the chart also uses a label, sorting, or baseline marker so that the message remains readable without color.

6.6 Dynamic Risk-by-Feature Chart

The most technically complex dashboard visual is the dynamic bar chart for *Average Predicted Churn Risk by Selected Feature*. In the Prediction page, a single-select feature slicer controls which business feature is shown, while the bar chart displays the categories inside that selected feature. We use a long-format table so one chart can switch between contract type, late payments, satisfaction, complaints, service calls, and the other selected features.

The slicer uses `feature_display`, not the technical `feature` field. The chart uses `category` on the category axis and `avg_churn_probability_pct` on the value axis. The value aggregation is **Average**, not **Sum**, because probabilities are not additive. This is why the bars in this chart are not expected to add up to 100%: each bar is a category-level average predicted probability, not a share of customers.

We also use the wording *Average Predicted Churn Risk* deliberately. It reminds the presenter that the chart compares average probability by category, not the distribution of customers across risk levels.

Table 13: Required long-format fields for the dynamic risk chart

Field	Role in the visual
feature	Technical feature name from the data pipeline.
feature_display	Readable business name used in the slicer.
category	Category shown on the bar-chart axis after a feature is selected.
customers	Number of scored customers in the selected category.
predicted_churn	Number of customers above the prediction threshold.
avg_churn_probability_pct	Average predicted churn probability in percent.

The chart is sorted in descending order so the highest-risk category appears first. Tooltips provide customer count, predicted churn count, and average predicted probability. The scope is also explicit: this dynamic risk chart is calculated from active non-churn customers scored by the model, while the separate high/medium/low risk summary on the model-evaluation page represents the test-period evaluation.

6.7 Page-Level Visualization Structure

We read the final visualization artifact as a compact story rather than a collection of independent charts:

1. **Overview:** shows total customers, churn count, non-churn count, baseline churn rate, and risk-level distribution.
2. **Churn drivers:** compares feature-group churn rates against the baseline and shows top model feature importance.
3. **Churn-only composition:** shows the share of churned customers by selected segment, while clearly stating that the denominator is churned customers only.
4. **Model evaluation:** shows AUC, accuracy, precision, recall, F1-score, confusion matrix, and a short interpretation of model limitations.
5. **Prediction and retention priority:** shows predicted churn customers, risk levels, and the features that define priority groups.

6.8 Final Dashboard Screenshots

The final reporting artifact contains three dashboard pages. Given the submission timeline, the group prioritized the techniques that most directly support the story: KPI cards summarize the problem size, bar charts compare segment risk, the feature slicer supports targeted exploration, and model outputs support retention prioritization.

6.9 Figure Quality Checklist

We used the following checklist when reading the final screenshots:

- The title states either the analytical question or the main insight.
- The denominator of every percentage or rate is explicit.
- Count views and churn-rate views are not mixed into one ambiguous metric.
- Churn-rate charts include the 9.92% baseline reference.

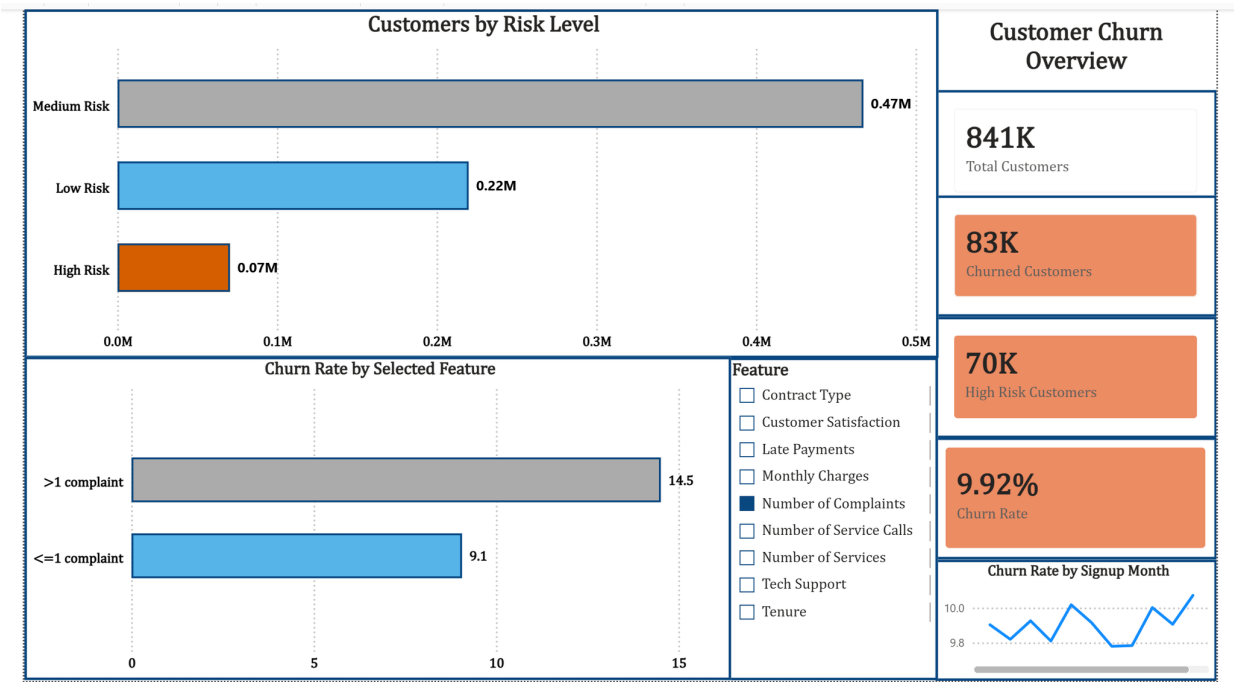


Figure 1: Final dashboard page 1: customer churn overview, KPI cards, risk-level distribution, selected-feature churn rate, and signup-month trend.

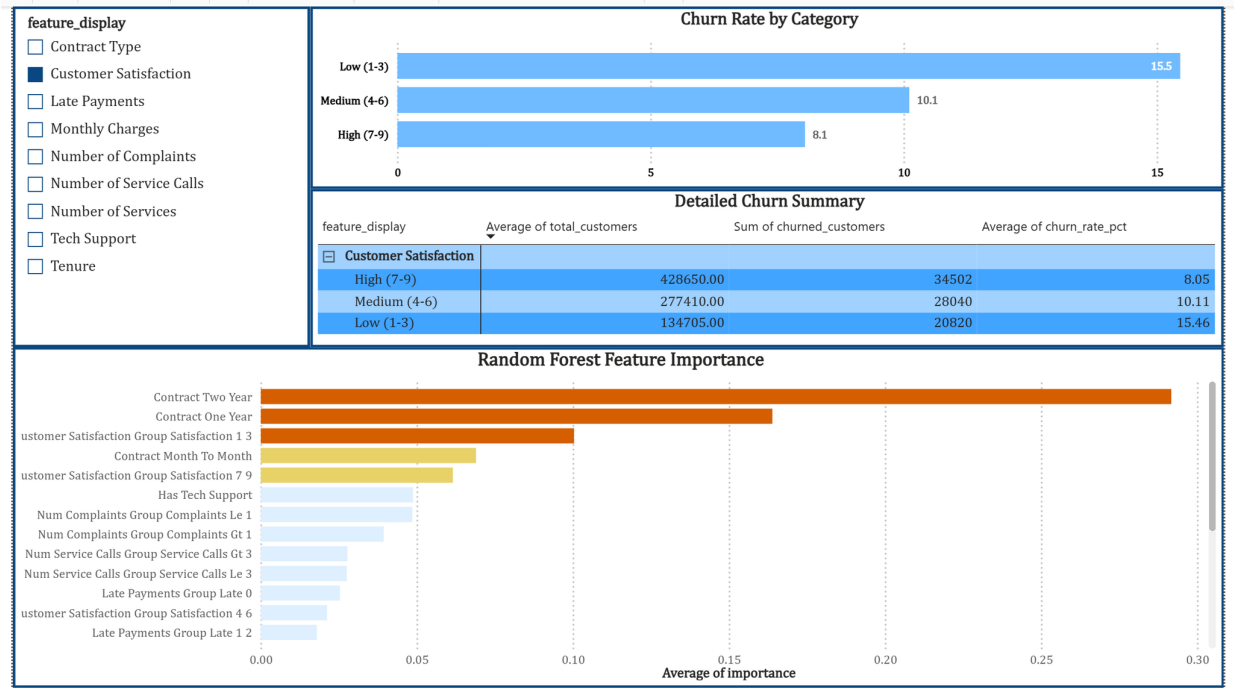


Figure 2: Final dashboard page 2: churn driver detail page with feature filter, churn-rate-by-category view, detailed summary table, and Random Forest feature importance.

- Category order is meaningful: ordinal groups keep natural order, while nominal groups are sorted by value only when ranking is intended.
- Colors are semantic and consistent across pages.
- The chart avoids 3D effects, unnecessary decoration, and hard-to-read legends.

7 Technique Application According to Course Guideline

7.1 Guideline Alignment

The course guideline at <https://datalab.vn/articles/2021-09/IT5425-quan-tri-du-lieu-va-truc-quan-hoa> asks each project to explain which visualization techniques are used and where they appear in the final artifact. We use the course slides and the distilled note file `Slide data viz/2024/notes/00_distilled_knowledge.md` as the reference, then connect the concepts directly to the three final dashboard pages.

Table 14: Course knowledge used in the final dashboard

Course knowledge	How we used it	Dashboard evidence
Visual encoding	We map categories to bar positions and quantitative values to bar length. Color is reserved for churn/risk emphasis.	Overview KPI page, selected-feature churn-rate chart, and Random Forest importance bars.
Graphical perception	We rely on common baselines, sorted categories, and preattentive orange highlights instead of dense legends.	Churn driver page and dynamic risk chart rank higher-risk categories first.
Table-data visualization	We aggregate customer-level rows into feature groups, rates, counts, and model summary tables.	Driver detail table, confusion matrix, split table, and risk-by-feature table.
Figure design	We avoid decorative 3D effects, keep direct labels, state denominators, and use a gray–orange–blue palette for accessibility.	All final screenshots use flat charts, KPI cards, labels, and consistent risk colors.
Interaction	We use slicers and view changes so one dashboard page can answer several feature-level questions.	Feature filters on the driver page and the single-select dynamic risk chart on the prediction page.
Storytelling	We order the dashboard from context to drivers and then to model-assisted retention priority.	Page 1 gives the churn baseline, page 2 explains drivers, and page 3 supports retention targeting.

7.2 Chapter 1: Overview of Data Visualization

Techniques / principles applied. We use visualization for both exploration and explanation. The EDA views help the group compare churned and non-churned customers, while the final dashboard explains which customer groups are above the baseline churn rate.

How applied in the dashboard. The overview page starts with total customers, churn count, churn rate, and risk distribution. This gives the audience a baseline before the report moves into feature-level churn

drivers and prediction outputs.

Notes / adjustments. The raw dataset is too large for direct row-level inspection during presentation, so the dashboard uses aggregated segment views and summary cards.

7.3 Chapter 2: Visual Models and Visual Encoding

Techniques / principles applied. The selected variables have different measurement levels: contract type is nominal, satisfaction and payment groups are ordinal/discrete, and monthly charges or usage are quantitative. We therefore use categorical axes for groups and bar length for counts, churn rates, feature importance, and predicted risk.

How applied in the dashboard. The selected-feature churn-rate chart maps feature categories to bars and within-category churn rate to bar length. The dynamic risk chart uses the same logic: the slicer selects a business feature, the axis shows its categories, and the bar length shows average predicted churn probability.

Notes / adjustments. We avoid stacking unrelated risk metrics into one visual. Churn rate, customer share, and average predicted probability are kept as separate metrics because they answer different questions.

7.4 Chapter 3: Graphical Perception

Techniques / principles applied. We use encodings that people compare accurately: position, length, ordering, and clear baselines. Color is used to guide attention, not to carry the whole meaning.

How applied in the dashboard. The driver page places feature-level charts and detail tables together, so the viewer can compare category risk and segment size in the same context. The dynamic risk chart sorts categories by average predicted churn probability, so the highest-risk category appears first.

Notes / adjustments. The gray–orange–blue palette helps draw attention to risk while staying more readable than red–green contrast. Labels, sorting, and chart titles remain available for viewers who cannot rely on color.

7.5 Chapter 4: Visualization for Table and Multidimensional Data

Techniques / principles applied. The project is mainly a table-data visualization problem. We transform customer-level records into grouped features, rates, counts, feature-importance tables, confusion matrices, and risk summaries.

How applied in the dashboard. The churn-driver page compares grouped categories against the overall churn baseline. The prediction page uses the long-format `RiskByFeature` table so the same bar chart can switch between contract type, satisfaction, complaints, service calls, late payments, and other selected features.

Notes / adjustments. Churn-rate bars and average-probability bars do not sum to 100%. Each bar is a within-category metric. Only composition charts, such as share of churned customers, are expected to sum to a whole.

7.6 Chapter 5 and Chapter 7: Graph and Map Visualization

Techniques / principles applied. Graph and map techniques are not central to this dataset because the available fields do not contain network edges or geographic locations.

How applied in the dashboard. The final dashboard therefore focuses on table-data visualization rather than adding graph layouts or maps that the data cannot support.

Notes / adjustments. If later data adds regions, stores, or customer relationships, this section can be extended with map encoding or graph-layout choices.

7.7 Chapter 6: Principles of Figure Design

Techniques / principles applied. We apply proportional ink, clear titles, readable labels, consistent colors, and simple layouts. The final pages avoid 3D effects and unnecessary decoration because the main task is comparison.

How applied in the dashboard. The overview page uses KPI cards for scale, the driver page uses bars and tables for segment comparison, and the prediction page combines model metrics with the dynamic risk chart. The gray–orange–blue palette gives a stable visual language: orange for warning or higher risk, blue for the main selected comparison, and gray for context.

Notes / adjustments. The dashboard still has room for typographic polish, but the report focuses on the design decisions that materially affect interpretation: denominator clarity, aggregation choice, sorting, and color accessibility.

7.8 Chapter 8: Interactive Visualization

Techniques / principles applied. The dashboard uses filtering, selection, and view changes. Interaction is used to support comparison rather than to hide the baseline or replace explanation.

How applied in the dashboard. The most advanced interaction is the Power BI dynamic risk chart. It uses a single-select slicer based on `RiskByFeature[feature_display]`; the bar chart then shows `RiskByFeature[category]` and the Average of `RiskByFeature[avg_churn_probability_pct]`. Selecting contract type changes the bars to month-to-month, one-year, and two-year contracts; selecting late payments changes the bars to late-payment groups.

Notes / adjustments. This chart is not a risk distribution. It compares average predicted churn probabilities, so the bars are not expected to add up to 100%. Using Average instead of Sum prevents probability values from being interpreted incorrectly.

7.9 Chapter 9: Storytelling with Data

Techniques / principles applied. The narrative moves from context to diagnosis and then to action. This matches the retention use case: managers first need the churn baseline, then the main drivers, and finally a way to prioritize customers.

How applied in the dashboard. The three final pages follow that order. Page 1 frames the overall churn problem, page 2 explains feature-level churn drivers, and page 3 connects model outputs with retention priority through predicted churn risk and the confusion matrix.

Notes / adjustments. The final wording avoids treating the model as a perfect classifier. The model supports prioritization, while the dashboard and report provide the business context needed to interpret the predictions.

8 Expected Insights and Recommendations

8.1 High-Priority Signals

The strongest expected signals are related to customer experience:

- Low customer satisfaction.
- High number of complaints.
- High number of service calls.

These features are directly actionable because they can guide the customer support and retention teams toward customers who may be experiencing service problems.

8.2 Moderate Signal

Late payments are expected to have a moderate positive relationship with churn. This feature should be interpreted together with satisfaction, complaint history, and service calls before it is used for retention targeting.

8.3 Weak Standalone Signals

Average monthly GB usage, days since last interaction, and credit score are expected to have weak standalone separation between churn and non-churn customers. They may still be useful as filters or as supporting variables in a multivariate model.

8.4 Prediction Workflow Insight

The Random Forest workflow in the group repository supports the EDA findings by using interpretable churn-related features such as contract, satisfaction, complaints, service calls, late payments, monthly charges, service count, and tech support. Because the churn class is imbalanced, the model is more suitable for ranking retention priority than for making a final yes/no business decision without human review.

8.5 Retention Recommendations

1. Prioritize customers with low satisfaction scores for proactive support.
2. Create a complaint escalation view for customers with multiple complaints.
3. Monitor repeated service calls as a sign of unresolved technical issues.
4. Combine late-payment signals with experience-related signals before launching retention campaigns.
5. Avoid drawing strong conclusions from usage volume or credit score alone.

8.6 Storytelling Flow for the Final Report

The final report narrative should move from context to action:

1. Establish why churn matters for a subscription-based telecom company.
2. State the baseline churn rate and total customer population.
3. Separate segment size from within-segment churn risk.
4. Highlight above-baseline customer groups using EDA evidence.
5. Connect the EDA findings with model evaluation and feature importance.
6. End with retention actions that are tied to customer experience variables.

This narrative avoids treating the model as the only result. The model supports prioritization, while the visualization explains which segments are actionable and why they deserve attention.

9 Limitations and Next Steps

9.1 Current Limitations

- Most dashboard interpretations still come from single-feature EDA, so the findings describe strong segment signals rather than full causal explanations.
- The model metrics require careful reading because the churn class is imbalanced; accuracy is less informative than recall, precision, F1-score, ROC AUC, and PR-AUC for this task.
- Several numeric variables are grouped for dashboard readability, which improves interpretation but can reduce predictive detail for the model.
- Some likely churn drivers are not available in the dataset, including detailed payment history, plan-change events, discount exposure, competitor offers, and support-ticket content.
- The final dashboard prioritizes the course techniques that best support this churn story. More optional views, such as small multiples or additional distribution plots, remain possible future extensions.
- Some visual polish remains possible, especially typography, page-level titles, and a final contrast audit for every chart.

9.2 Next Steps

1. Re-check all churn-rate values against the latest cleaned dataset.
2. Add more multivariate analysis after the single-feature EDA is stable.

3. Review Random Forest feature importance and high-risk customer examples.
4. Test raw numeric features together with grouped features and compare Random Forest with XGBoost, LightGBM, or CatBoost.
5. Add PR-AUC, threshold analysis, and class-imbalance handling to the modeling evaluation if the predictive workflow is extended.
6. If more time is available, refine chart titles, verify color contrast, and audit churn-rate denominators directly in the dashboard artifact.
7. Add more final insight paragraphs after the group completes the oral presentation narrative.

10 Conclusion

This report connects the churn dataset, EDA workflow, feature grouping, Random Forest outputs, and final Power BI dashboard into one analysis narrative. The main interpretation separates distribution from risk: customer counts explain segment size, while churn rates and average predicted probabilities explain which groups need attention. The final dashboard applies course concepts from visual encoding, graphical perception, table-data visualization, figure design, interaction, and storytelling. In the current results, customer-experience features such as satisfaction, complaints, service calls, late payments, and contract type provide the clearest signals for retention prioritization, while the model supports ranking customers rather than replacing business judgement.

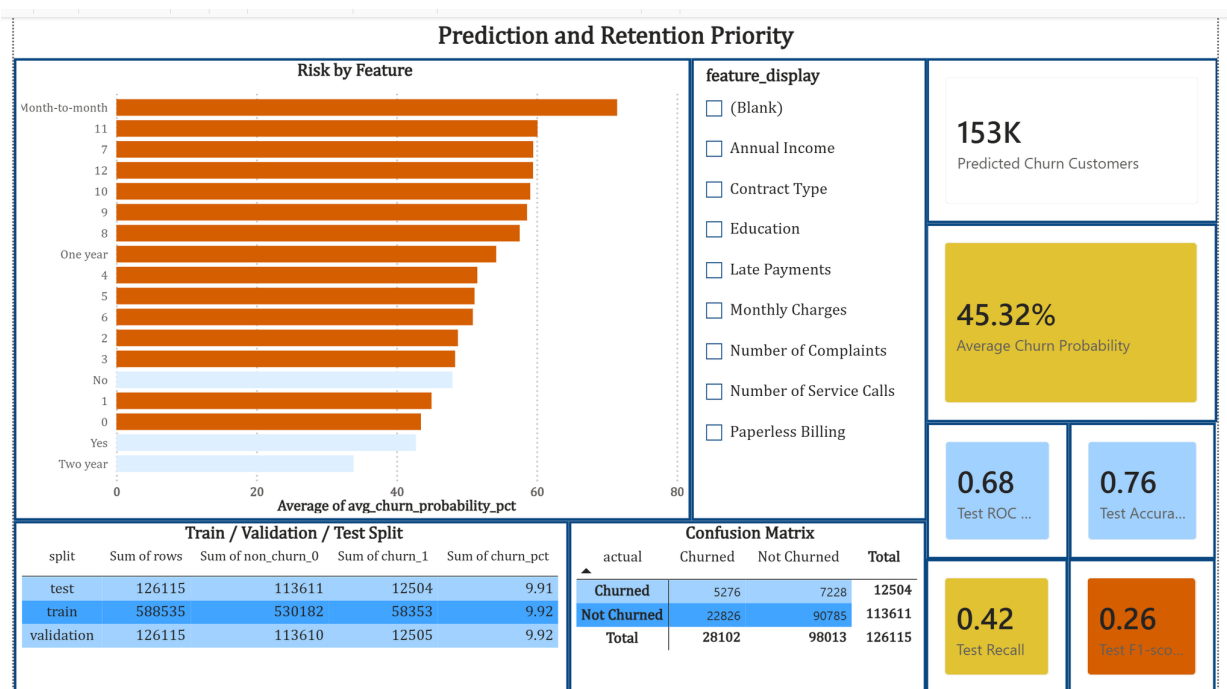


Figure 3: Final dashboard page 3: prediction and retention priority page with risk-by-feature view, predicted churn KPIs, train/validation/test split, and confusion matrix.